

Infosys Springboard Internship Project

SUPPLY CHAIN VISIBILITY & LOGISTICS

Data Preprocessing and Power BI Visualization

Prepared by

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DECLARATION

I hereby declare that this report presents my work as part of the Infosys Internship Project on Supply Chain Visibility. My contribution was the visualization stage under the Supplier Performance Scorecard topic. After the project team collected the raw data and completed preprocessing, I used the cleaned dataset to develop two Power BI dashboards: an overview dashboard and a detailed supplier scorecard dashboard.

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Date: 07:009:2026

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ACKNOWLEDGEMENT

I sincerely thank Infosys, my institution, faculty members and internship mentors for giving me the opportunity to work on the Supply Chain Visibility project. I am grateful for the guidance provided throughout the project.

I also thank my project team members for their cooperation in collecting the raw data, preprocessing the dataset, discussing the project requirements and completing the overall work. This project gave me practical experience in data cleaning and Power BI dashboard development.

ABSTRACT

The Infosys Internship Project on Supply Chain Visibility follows a data-analysis workflow beginning with raw data collection, followed by data preprocessing and finally visualization. The supply-chain dataset contains information related to orders, customers, products, suppliers, warehouses, delivery dates, delay days, transportation cost, order value and shipping mode.

After the raw data was collected, the project team performed preprocessing to improve the quality and consistency of the dataset. The preprocessing included checking missing values and duplicates, standardizing column names and text, handling date values, calculating delivery delay, correcting delivery status and filling selected missing numerical and date values.

After preprocessing, I worked on the visualization part of the project under the topic Supplier Performance Scorecard. I created two dashboards using the cleaned dataset. The first dashboard provides an overview of supplier and supply-chain performance, while the second dashboard presents a supplier scorecard overview using KPIs, supplier comparisons, warehouse location, shipping modes, transportation cost and monthly trends.

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1. INTRODUCTION

Supply chain visibility is the ability to understand and monitor important activities across the supply chain. It includes information about orders, suppliers, warehouses, transportation, delivery dates and costs. Data visualization helps convert these records into information that can be understood quickly.

The Infosys Internship Project on Supply Chain Visibility applies this approach in three main stages: raw data collection, data preprocessing and data visualization. The final visualization stage uses Power BI to create dashboards for understanding supply-chain performance.

My assigned part was the Supplier Performance Scorecard topic. I created two dashboards from the cleaned dataset and focused on supplier performance, order value, transportation cost, delivery status, shipping mode, warehouse location and monthly trends.

2. INFOSYS INTERNSHIP PROJECT OVERVIEW

The overall internship project is focused on Supply Chain Visibility. The project team first collected the raw supply-chain dataset. The data was then examined and preprocessed so that it could be used for analysis. After cleaning, the dataset was visualized using Power BI.

2.1 Overall project workflow

1. Collection of raw supply-chain data
2. Inspection and understanding of the raw dataset
3. Data preprocessing and cleaning
4. Preparation of the cleaned dataset
5. Visualization using Power BI
6. Development of dashboards according to assigned topics
7. Analysis and presentation of dashboard results

2.2 Individual contribution

I worked on the visualization stage after the preprocessing was completed. My topic was Supplier Performance Scorecard. I created two dashboards under this same topic rather than treating them as separate project topics.

3. SUPPLY CHAIN VISIBILITY

The dataset provides a combined view of operational information such as orders, suppliers, warehouses, dates, delivery performance, transportation and order value. Such a dataset can be used to understand how suppliers contribute to overall order value and how delivery and transportation performance varies.

3.1 Importance of visualization

- Summarizes large amounts of data in a simple format.
- Helps compare suppliers and warehouses.
- Shows delivery performance using status and delay measures.
- Provides visibility into transportation modes and costs.
- Shows monthly changes in order activity and order value.
- Supports quick interpretation through KPI cards and charts.

4. RAW DATA COLLECTION

The first stage of the project was collecting the raw supply-chain data. The supplied raw file contains 1,000 records and 14 columns. The dataset includes information about orders, customers, products, suppliers, warehouses, dates, delivery performance, transportation cost, order value and shipping mode.

4.1 Raw dataset fields

Order_ID • Customer_ID

Product_ID • Supplier

Warehouse • Order_Date

Dispatch_Date • Delivery_Date

Expected_Delivery_Date • Delivery_Status

Delay_Days • Shipping_Mode

Transportation_Cost • Order_Value

The raw dataset was not directly used for the final dashboards. It first went through preprocessing so that the data could be used more consistently for visualization.

5. DATA PREPROCESSING

Data preprocessing was performed after raw data collection and before visualization. The purpose was to improve data quality, standardize fields and prepare the dataset for Power BI.

5.1 Data inspection

The dataset was loaded into Python using Pandas and inspected using row/column information, sample records, column names, data types and descriptive statistics.

5.2 Missing-value check

Missing values were identified in relevant columns and handled using the preprocessing logic implemented in the project.

5.3 Duplicate check

Duplicate records were checked. The preprocessing notebook recorded zero duplicate rows.

5.4 Column-name standardization

5.5 Text cleaning

Extra spaces were removed from identifier and text fields to make values consistent.

5.6 Date handling

Order Date, Dispatch Date and Expected Delivery Date were checked and missing values were filled using the project's date-processing logic.

5.7 Delay calculation

Delay Days was recalculated using the difference between Delivery Date and Expected Delivery Date.

5.8 Delivery-status correction

Status was aligned with Delay Days: negative values indicate Early, zero indicates On Time and positive values indicate Delayed.

5.9 Numerical-value handling

Missing Transportation Cost and Order Value values were completed using the project's median-based approach.

5.10 Cleaned-data export

The processed data was saved as a cleaned dataset for further analysis and visualization.

5.11 Why preprocessing was important

Without preprocessing, missing values, inconsistent text and incorrect status information could affect Power BI calculations and visualizations. Cleaning the dataset therefore provided a better base for creating supplier KPIs and charts.

6. CLEANED DATASET

The supplied cleaned dataset contains 1,000 rows and 14 columns. It was used as the main input for the Power BI visualization stage.

Field	Use in dashboard
Order ID	Order counting and identification
Customer ID	Customer-level identification
Product ID	Product-level identification
Supplier	Supplier comparison and ranking
Warehouse	Location and warehouse analysis
Order Date	Time-based analysis
Dispatch Date	Operational date analysis
EDD	Expected delivery comparison
Delivery Date	Actual delivery comparison
Delay Days	Delay KPI and delivery analysis
Status	Early / On Time / Delayed comparison
Transportation Cost	Cost KPI and supplier cost analysis
Order Value	Value KPI and supplier value analysis
Shipping Mode	Sea / Road / Rail / Air analysis

The cleaned dataset was then imported into Power BI for visualization.

7. DATA VISUALIZATION

After preprocessing, the next stage was visualization. Power BI was used to transform the cleaned dataset into interactive dashboards. The visualization stage was focused on presenting the data in a way that makes supplier and supply-chain performance easy to understand.

7.1 Visualization process

8. Import the cleaned dataset into Power BI.
9. Check the fields and their data types.
10. Create measures/KPIs required for the scorecard.
11. Select suitable visual types for each analysis.
12. Arrange the visuals into a clear dashboard layout.
13. Add titles and labels for easy interpretation.
14. Review the final dashboard for readability and usefulness.

8. MY CONTRIBUTION – SUPPLIER PERFORMANCE SCORECARD

My contribution was limited to the visualization part under the topic Supplier Performance Scorecard. I created two dashboards using the same cleaned dataset.

Dashboard	Purpose
Dashboard 1 – Overview	Provides a broad overview of supplier and supply-chain performance using KPIs and summary visuals.
Dashboard 2 – Supplier Performance Scorecard	Provides a more detailed supplier scorecard view with supplier, warehouse, shipping and trend analysis.

The two dashboards are different views of the same topic and are not separate project topics.

9. DASHBOARD 1 – OVERVIEW DASHBOARD

The first dashboard provides an overall view of supplier performance. It combines KPI cards with a warehouse map, shipping-mode distribution, supplier order-value comparison, transportation cost and monthly trend analysis.



Figure 1: Supplier Scorecard – Overview Dashboard

9.1 KPI cards

KPI	Displayed value	Purpose
Total Order Value	30.96M	Shows total order value.
Total Suppliers	1K	Shows the supplier count displayed in the dashboard.
Transportation Cost	1.30M	Shows total transportation cost.
Delay-related measure	≈2K	The visible card is based on Sum of Delay_Days.

9.2 Warehouse Location

The map provides a geographic view of warehouse locations and helps connect supply-chain activity with physical locations.

9.3 Count of Shipping Mode

The donut chart shows the distribution of orders across shipping modes such as Sea, Road, Rail and Air.

9.4 Order Value by Supplier

The horizontal bar chart compares suppliers according to their total order value and provides a supplier ranking.

9.5 Transportation Cost by Supplier

This bar chart compares transportation costs across suppliers and helps identify differences in logistics expenditure.

9.6 Monthly order trend

The line chart shows changes in order activity/order-value-related information across the months.

10. DASHBOARD 2 – SUPPLIER PERFORMANCE SCORECARD

The second dashboard is the detailed Supplier Performance Scorecard. It focuses on operational performance through order count, order value, transportation cost, delay, supplier status, warehouse status, shipping mode and monthly trends.

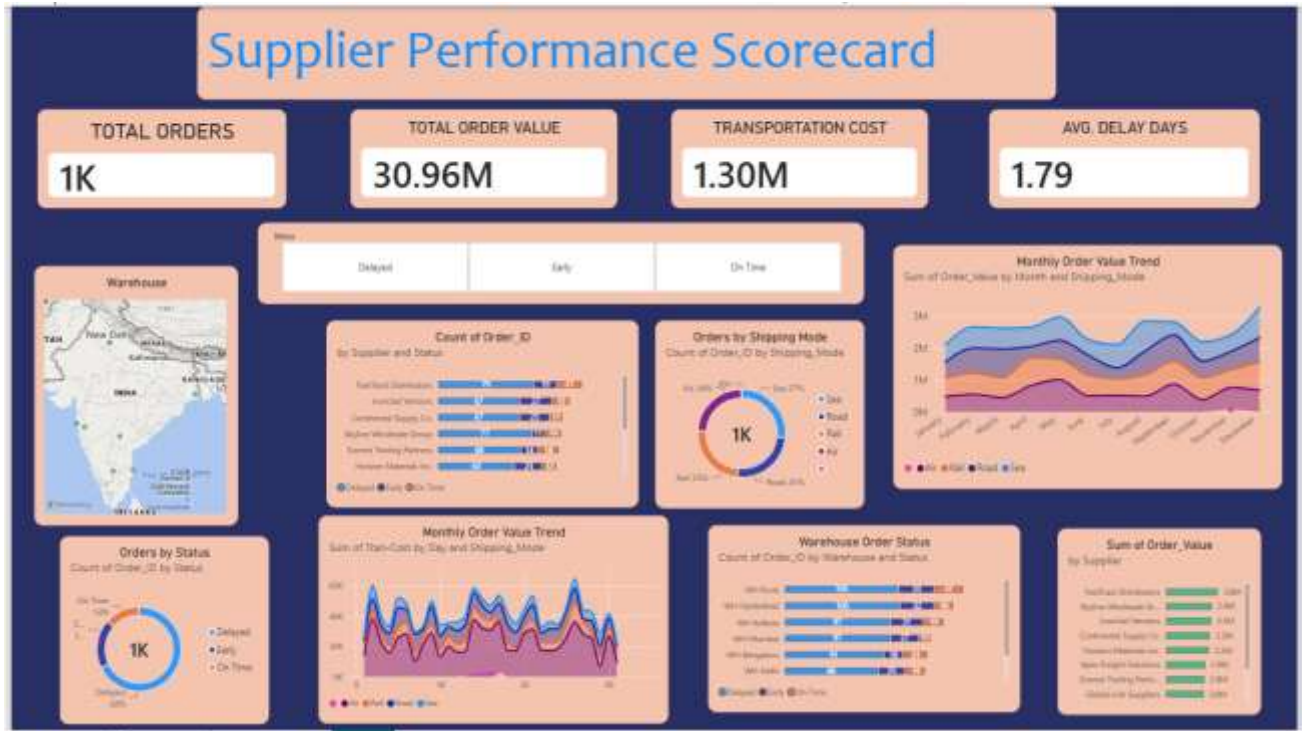


Figure 2: Supplier Performance Scorecard Dashboard

10.1 KPI cards

KPI	Displayed value	Purpose
Total Orders	1K	Shows the total number of orders.
Total Order Value	30.96M	Shows the aggregated order value.
Transportation Cost	1.30M	Shows the total transportation cost.
Avg. Delay Days	1.79	Shows the displayed average delay KPI.

10.2 Warehouse map

The warehouse map provides a geographic view of warehouse locations.

10.3 Supplier and delivery status

The stacked bar chart compares supplier order counts by Delayed, Early and On Time status. It helps identify differences in supplier delivery performance.

10.4 Orders by shipping mode

The donut chart shows the distribution of orders by shipping mode. Sea, Road, Rail and Air are represented.

10.5 Monthly order value trend

The monthly trend chart shows how order value changes across the year, allowing higher and lower activity periods to be compared.

10.6 Warehouse order status

This chart compares order status across warehouses and helps identify the distribution of delayed, early and on-time orders.

10.7 Supplier order value

The supplier ranking chart compares the total order value associated with each supplier.

11. DASHBOARD ANALYSIS

11.1 Order performance

The dashboards provide a high-level view of approximately 1K orders and a total order value of 30.96M. This gives an immediate understanding of the overall scale represented by the dataset.

11.2 Supplier performance

Supplier-wise charts allow comparison of order value and delivery status. This can help identify suppliers with higher order contribution and differences in delivery performance.

11.3 Transportation performance

Transportation cost is shown as 1.30M in the dashboards. Supplier-wise transportation cost visuals provide an additional view of logistics expenditure.

11.4 Delivery performance

The cleaned dataset uses Delay Days to classify delivery status. Early, On Time and Delayed categories can therefore be compared in the dashboard.

11.5 Warehouse performance

The warehouse map and warehouse-status chart provide both geographic and operational views of warehouse activity.

11.6 Monthly trends

Monthly charts make it possible to observe changes in order activity and order value throughout the year.

12. KEY FINDINGS

- The project follows the sequence raw data collection → preprocessing → cleaned dataset → visualization.
- The cleaned dataset contains 1,000 records and 14 fields.
- The dashboard work is one part of the larger Infosys Internship Project on Supply Chain Visibility.
- My individual topic was Supplier Performance Scorecard.
- I created two dashboards under the same Supplier Performance Scorecard topic.
- The dashboards show 1K total orders and 30.96M total order value.
- Transportation cost is shown as 1.30M.
- The detailed dashboard displays 1.79 as its average delay KPI.
- Supplier order-value charts allow suppliers to be ranked and compared.
- Shipping-mode visuals provide a distribution of Sea, Road, Rail and Air orders.
- Warehouse visuals provide location and delivery-status information.
- Monthly trend charts support comparison across months.

13. LIMITATIONS AND DATA QUALITY NOTES

The report is based on the supplied raw dataset, cleaned dataset and the two completed dashboard screenshots. The dashboards are intended for academic/internship analysis.

The preprocessing notebook shows some remaining non-null counts below 1,000 for selected text fields in its final information output. These fields should be reviewed if the dataset is later used for production reporting.

The second dashboard's card titled "Average Delay Days" visibly uses "Sum of Delay_Days" as its subtitle. If an exact average is required, the Power BI measure should be changed to Average of Delay_Days. The detailed dashboard's displayed 1.79 KPI is treated as the value shown in the supplied dashboard.

14. CONCLUSION

The Infosys Internship Project on Supply Chain Visibility demonstrates the complete flow from raw data collection to preprocessing and visualization. The team first collected the raw dataset and prepared it through data-cleaning activities. The cleaned dataset then became the basis for Power BI analysis.

My contribution was the visualization stage under the Supplier Performance Scorecard topic. I created two dashboards using the cleaned dataset: an Overview Dashboard and a detailed Supplier Performance Scorecard Dashboard. These dashboards present supplier performance, order value, transportation cost, delivery status, shipping modes, warehouse locations and monthly trends.

The project provided practical experience in converting cleaned data into meaningful visual information and demonstrated how Power BI can be used to improve supply-chain visibility.

15. FUTURE SCOPE

- Add slicers for supplier, warehouse, shipping mode and month.
- Create drill-through pages for individual suppliers.
- Develop a supplier score based on delivery, cost and order value.
- Add alerts for repeated delivery delays.
- Build predictive models for delivery-delay risk.
- Forecast future order value and supply-chain activity.
- Add route and carrier performance analysis.
- Publish the dashboard for regular monitoring.

16. REFERENCES

- Pandas documentation used for data preprocessing.
- Python documentation.
- NumPy documentation.
- Microsoft Power BI documentation.
- Google Colab / Jupyter Notebook resources.
- Supply Chain Visibility raw dataset supplied for the internship project.
- Supply Chain Visibility cleaned dataset supplied for the internship project.

APPENDIX – MY PROJECT CONTRIBUTION SUMMARY

Stage	Team / Project activity	My contribution
1	Raw data collection	Part of the overall project workflow.
2	Data preprocessing	Used the cleaned output prepared for visualization.
3	Data visualization	Primary individual contribution.
4	Topic	Supplier Performance Scorecard.
5	Dashboard 1	Created Overview Dashboard.
6	Dashboard 2	Created Supplier Performance Scorecard Dashboard.
7	Analysis	Explained KPIs, supplier, warehouse, shipping and trend visuals.

Final statement: The two dashboards documented in this report are my visualization work under the Supplier Performance Scorecard topic of the Infosys Internship Project on Supply Chain Visibility.